🔶 SECTION 3: Density & Unit Cell Calculations (Q41–70)

Q	Question	Answer	Explanation
41	Density of a solid is given by:	C) ρ = ZM / (a^3 N_A)	ρ = mass/unit cell ÷ volume; Z = atoms/unit cell, M = molar mass, a^3 = volume, N_A = Avogadro's number
42	Number of atoms per unit cell (FCC)	C) 4	8 corners × 1/8 + 6 faces × 1/2 = 4 atoms
43	Number of atoms per unit cell (BCC)	B) 2	8 corners × 1/8 + 1 center = 2 atoms
44	Number of atoms per unit cell (SC)	A) 1	8 corners × 1/8 = 1 atom
45	Density of Cu, r = 0.128 nm, FCC	A) 8.92 g/cm^3	Use ρ = ZM / (a^3 N_A), a = 2√2 r
46	Number of formula units in NaCl	B) 4	FCC lattice: 4 Na^+ + 4 Cl^- per unit cell
47	SC unit cell, a = 4 Å, M = 20 g/mol	B) 4 g/cm^3	ρ = ZM / (a^3 N_A), Z=1 for SC
48	Atomic packing factor (APF)	A) Volume of atoms / Volume of unit cell	Measures fraction of cell volume occupied by atoms
49	In FCC, r vs a	B) a = 2√2 r	Face diagonal = 4r → a√2 = 4r
50	Density of BCC metal	A) Higher than SC	BCC has more atoms per unit cell than SC, density increases
51	Na^+ ions in NaCl unit cell	C) 4	4 Na^+ ions in FCC lattice

Q	Question	Answer	Explanation
52	Cl^- ions in NaCl unit cell	C) 4	4 Cl^- ions in FCC lattice
53	Edge length doubles, density	C) Reduces to one-fourth	$\rho \propto 1/a^3$
54	Mass of atoms per unit cell	A) Z × atomic mass / N_A	Z = number of atoms/unit cell
55	Lattice constant from density	A) $a = (Z M / \rho N_A)^{(1/3)}$	Derived from ρ formula
56	Radius relation, octahedral void	A) $R_o = 0.414 r$	Standard relation in close packing
57	Radius relation, tetrahedral void	C) $R_t = 0.225 r$	Tetrahedral void smaller than octahedral
58	Tetrahedral voids per atom in FCC	B) 2	8 tetrahedral voids / 4 atoms = 2 per atom
59	Octahedral voids per atom in FCC	A) 1	4 octahedral voids / 4 atoms = 1 per atom
60	Packing efficiency ∝	C) Volume occupied by atoms	More volume occupied → higher packing efficiency
61	Density increases when	B) Number of atoms increases	More atoms/unit cell → higher density
62	HCP structure, atoms/unit cell	A) 6 atoms	Standard HCP unit cell contains 6 atoms
63	Ionic solids density depends on	D) All of the above	Depends on mass, number of ions, and edge length
64	SC structure lowest density	D) All of the above	Low packing, large voids, fewer atoms/unit cell

Q	Question	Answer	Explanation
65	Radius of ion in octahedral void	C) Both A & B	Depends on host atomic radius and CN
66	Tetrahedral void size	B) Smaller than octahedral	Standard geometric relation
67	Highest packing efficiency	C) FCC	FCC and HCP have 74% packing efficiency
68	Radius of FCC atom, a = 4.05 Å	D) 1.43 Å	$a = 2\sqrt{2}\, r \rightarrow r = a/(2\sqrt{2}) \approx 1.43$ Å
69	Formula units in CsCl	B) 2	CsCl is BCC-type structure: 1 Cs + 1 Cl in unit cell × 2
70	CsCl structure	B) BCC	Cs^+ at center, Cl^- at corners (or vice versa)

🔶 SECTION 4: Imperfections in Solids (Q71–90)

Q	Question	Answer	Explanation
71	Schottky defect	A) Vacancy defect	Missing cations and anions, preserves neutrality
72	Frenkel defect	C) Both A & B	Cation moves to interstitial site (vacancy + interstitial)
73	Colour of NaCl changes	C) F-centres	Electron trapped at anion vacancy absorbs light
74	Effect of Schottky defect on density	B) Decreases	Vacancies reduce mass/unit volume
75	Effect of Frenkel defect on density	C) No effect	Atom displaced but total mass unchanged
76	Edge dislocation affects	B) Mechanical properties	Slip and deformation occur along dislocations
77	Screw dislocation	B) Line defect	One-dimensional defect along a line

Q	Question	Answer	Explanation
78	Colour centres formed due to	A) Missing anions	Vacancies trap electrons → colour centres
79	Intrinsic defects caused by	B) Thermal agitation	Occur naturally due to temperature
80	Extrinsic defects caused by	B) Impurities	Caused by foreign atoms/ions
81	Defect common in AgCl	B) Frenkel	Ag^+ cations small → move to interstitials easily
82	Defects increase with	A) Temperature	Thermal agitation increases vacancy/interstitials
83	Grain boundary	B) Surface defect	Two grains meet → planar defect
84	Stacking fault	B) Surface defect	Local deviation from stacking sequence
85	Dislocations make metals	A) Harder	Interaction between dislocations → strengthening
86	Vacancy in ionic solids	A) Reduces density	Missing ions → lower mass/unit volume
87	Defects affect	D) All of the above	Mechanical, electrical, diffusion properties affected
88	Frenkel defect common in	B) AgCl	Small cation → interstitial movement
89	Schottky defect common in	A) NaCl	Equal missing cations/anions
90	F-centres responsible for	A) Colour	Trapped electron absorbs visible light

🔶 SECTION 5: Electrical & Magnetic Properties (Q91–120)

Q	Question	Answer	Explanation
91	Conductivity of metals	A) High	Metals have free electrons
92	Diamond	B) Insulator	No free electrons in sp^3 network
93	Si	B) Semiconductor	Electrical conductivity intermediate
94	Intrinsic semiconductor conductivity increases with	A) Temperature	Thermal excitation creates e^--h^+ pairs
95	N-type semiconductor	A) Extra electrons	Doping with pentavalent atom
96	P-type semiconductor	B) Extra holes	Doping with trivalent atom
97	Diamagnetic materials	A) Weakly repelled	No unpaired electrons
98	Paramagnetic materials	A) Weakly attracted	Unpaired electrons respond to magnetic field
99	Ferromagnetic materials	A) Strongly attracted	Parallel alignment of magnetic moments
100	Curie temperature	A) Temp at which ferromagnet → paramagnetic	Magnetic ordering lost above T_C
101	Superconductors have	A) Zero resistance	Perfect conductors below T_c
102	Meissner effect	A) Expulsion of magnetic field	Perfect diamagnetism in superconductors
103	Hall effect used to determine	A) Type of semiconductor	Sign of charge carriers determined
104	Piezoelectric solids generate	A) Voltage on stress	Mechanical stress → electric polarization
105	Examples of piezoelectric solids	A) Quartz, Rochelle salt	Well-known piezoelectric materials

Q	Question	Answer	Explanation
106	Magnetic domains are	A) Regions of aligned magnetic moments	Ferromagnetic alignment at microscopic scale
107	Paramagnetic susceptibility	A) Positive	Attracted weakly to magnetic field
108	Diamagnetic susceptibility	B) Negative	Repelled weakly from magnetic field
109	Conductivity of semiconductors increases with	C) Both	Doping or temperature increases carrier density
110	Intrinsic semiconductors	A) Pure	Pure semiconductor without doping
111	Extrinsic semiconductors	A) Doped	Doping with donor/acceptor atoms
112	Example of ferromagnetic material	A) Fe	Fe, Co, Ni are ferromagnetic
113	Example of paramagnetic material	A) Al	Al is paramagnetic, weak attraction
114	Example of diamagnetic material	A) Cu	All electrons paired → repelled by magnetic field
115	Conductivity of metals decreases with	A) Temperature increase	Electron scattering increases at high T
116	Ferromagnetic materials retain magnetization	A) Yes	Magnetic domains aligned → retain magnetization
117	Semiconductors are	C) Intermediate conductivity	Conductivity between metals and insulators
118	Colour centres occur due to	A) Missing anions	Electron trapped at anion vacancy
119	Superconductivity destroyed above	A) Critical temperature	Resistance appears above T_c

Q	Question	Answer	Explanation
120	Magnetic hysteresis curve is for	A) Ferromagnetic materials	Shows magnetization vs applied field